

Cognitive Aspects

Assistive Devices for those on the Spectrum

This week's assignment will be a further extension on last week's as we develop the idea for the proposed device while analyzing different approaches and technique of cognitive aspects that we can adopt in order to improve and augment the way of life for people who are on the spectrum. Most importantly the significance of Cognitive Aspects in the area of Assistive Devices is large, as the users of these devices are often physically or mentally challenged. This can make it challenging for the developers to accommodate the complications when designing the interface of the device. Below, we will discuss every area of Cognitive Aspect and ways that we can implement them with the proposed device.

1. **Attention:** For people who are on the spectrum they are often impatient and easily irritable. Therefore, it is imperative that we design the interface in such a way that it is more practical than “pretty”. The various salient and key features must be easily available without much complications thus providing the users a comfortable and hassle-free experience
2. **Perception:** It is important that we design a device that is easily perceptible, as for an assistive device for GPS location. It is very likely that there will be various forms of image and text available. Hence, we need to ensure that the media being used is clear and comprehensible to avoid any confusion.
3. **Memory:** As we have discussed that the target group for this product are mentally/physically challenged. It is important that we provide them with tools such as Recognition which will be personalized to every use and enhance the user experience

4. **Learning:** It is an extension to the previous point, in that the software that is implemented in the device must be able to initially provide guidelines and act as a stepping stone while in parallel learning about the behavior, preferences and other crucial information about the user so that it can predict and automatically choose the most appropriate action for the user

5. **Reading, Speaking and Listening:** It is almost mandatory that any Assistive Device have a voice feature which can read and listen to audio input as well as provide an audio output. Hence, it is crucial that the voice being used is clear and crisp in order to improve comprehension